ECE 520 Final Project Proposal: Real - Time Signal Detection

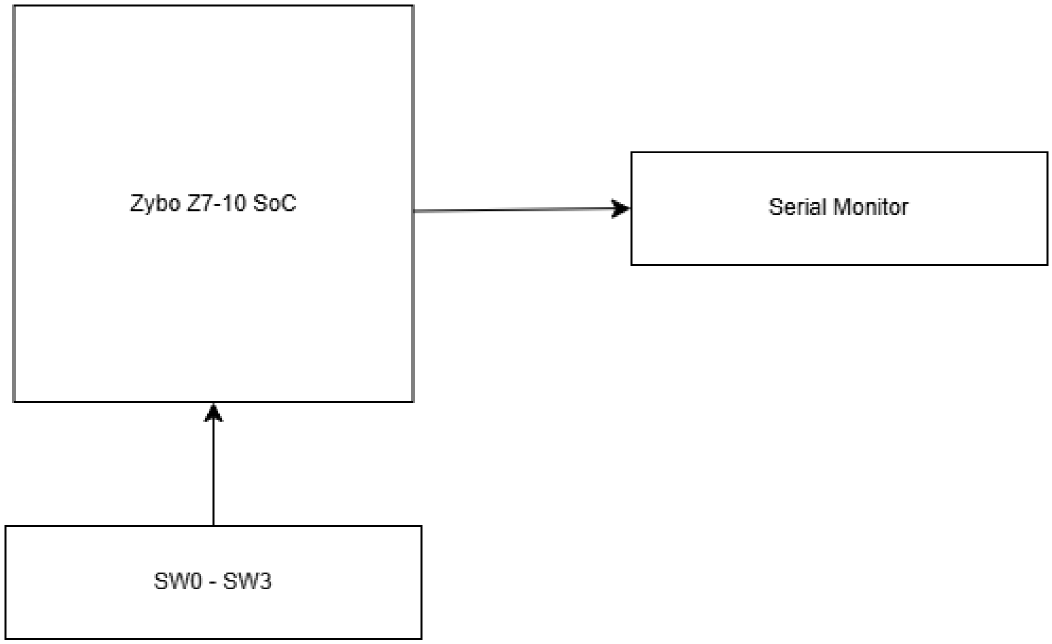
Objective

The objective of this project is to design and implement a real-time signal detection system on the Zybo Z7-10 board. The system will detect the presence of a signal, estimate its frequency using zero-crossing techniques, and measure its amplitude. The design will use both the programmable logic and processing system to demonstrate hardware/software co-design. Additionally, on-board switches will be used to dynamically select between multiple simulated input frequencies, allowing real-time validation of the detection algorithm.

Background and Methodology

This project applies key FPGA and SoC design concepts using the Zybo Z7-10, specifically leveraging the integration of programmable logic and an embedded processing system. This design utilizes hardware/software co-design, where computationally intensive, real-time signal processing is implemented in the FPGA fabric, while higher-level control and communication are handled by the ARM processor. The FPGA concepts applied in this project include synchronous digital design, finite state machines (FSMs), counters, and real-time data processing pipelines. The project also incorporates AXI-based communication between the PL and PS, demonstrating standard SoC interfacing techniques. On the software side, embedded C will be used to process results and communicate with a host system via UART. The expected outcome of this project is a functional real-time signal detection system capable of identifying the presence of a signal ,estimating its frequency, and measuring its amplitude. The system will respond dynamically to user inputs via on-board switches, allowing different signal frequencies to be generated and correctly identified.

Block Diagram



Pinout Plan

<u>Inputs: SW0 - SW3</u>
Used to select between different simulated signal frequencies. Connected to PL as digital inputs.

<u>Output: UART</u>
Used to display signal detection status, estimated frequency, and peak amplitude.

Components Used

- Zybo Z7-10
- USB Cable

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